

E1: Structure of graphs built by WUKONG, LightRAG and GraphRAG

UDA-Bench NBA, experiment report

October 9, 2026

1 Goal

We build a graph from the same corpus with three systems, using the same language model, and measure how the resulting graphs can be navigated and queried. The statistics are chosen so that they can be read without knowing the domain: size, node and relation vocabularies, edge direction and properties. Recall against the benchmark’s gold tables is reported only as a control, to check that the three graphs contain comparable information.

2 Setup

Corpus and gold. The NBA dataset of UDA-Bench: 216 English Wikipedia articles, 141 about players, 30 about teams, 29 about cities and 16 about team owners. The gold tables (player, team, city, manager) have one row per article; the row ID is the document number.

Systems. All three use `gpt-5.6-luna` with low reasoning effort and `text-embedding-3-small`.

- **WUKONG**, engine at commit `99d591e`, with the extraction schema in `workspace-v1/`: four entity types (Player, Team, City, Owner) with typed fields, and three relationship types (PlaysFor, with a `tenure` field; LocatedIn; Owns, with a `since_year` field). Chunks of 800 tokens with 120 overlap; export to Neo4j CSV.
- **LightRAG** 1.5.7, default extraction, chunks of 1200 tokens with 100 overlap, 16 concurrent LLM calls (`lightrag_build.py`).
- **GraphRAG** 3.2.0, standard indexing with its default entity types (organization, person, geo, event), one glean, chunks of 1200 tokens with 100 overlap, default prompts (`graphrag-config/`).

Schema hygiene. The first version of the WUKONG schema (`workspace/`) used example values taken from the corpus, including gold answers (two players’ birth dates, two cities’ states, a team’s owner, and subject names). We replaced every example by a value that is neither a gold cell nor the name of a subject entity, leaving descriptions, instructions, formats and options unchanged (`workspace-v1/`, built by `make_schema_variants.py`, which rejects any schema that fails this check). The reported WUKONG graph is built with the clean schema. Two conventions of the gold tables are used on purpose as target-schema knowledge: the closed vocabulary {Frontcourt, Backcourt} for a player’s position, and the date format YYYY/M/D. The baselines receive no schema.

Build cost. Table 1. Output tokens include reasoning tokens. Runs were executed concurrently with other runs, so wall times are indicative.

	LLM calls	Input tokens	Output tokens	Wall time
WUKONG	4,463	6.5M	1.1M	18 min
LightRAG	6,341	17.3M	7.8M	92 min
GraphRAG	24,965	35.3M	15.1M	101 min

Table 1: Cost of building each graph over the 216 documents. GraphRAG’s calls are 3,362 graph extractions, 16,511 description summaries and 5,092 community reports; it also used 8.2M embedding tokens.

3 Measures

Graph. For WUKONG we count domain nodes and edges; the provenance nodes and edges that link every object to its source chunk and document are left out. For LightRAG we read its GraphML graph; for GraphRAG its entity and relationship tables.

Relation vocabulary, own labels. The label of an edge is what the system stores: the relationship type in WUKONG, the set of keywords in LightRAG. GraphRAG stores no relation label, only a free-text description and a weight. For a labeling we report the number of distinct labels, the fraction used by a single edge, the number of most frequent labels needed to cover 50% and 90% of the edges, and the mean number of distinct labels per pair of node types (over pairs with at least 100 edges).

Relation vocabulary, derived predicates. To measure LightRAG and GraphRAG in the same way, and to give GraphRAG a label at all, an LLM (same model and effort as the builds) reads the description of every edge, together with the names of its two endpoints, and names the main relation as a short predicate in lower snake case, 1–3 words, present tense, without dates, numbers or names, and states which endpoint is its subject (`derive_labels.py`; the prompt’s examples come from an unrelated domain). Edges are labeled independently, 100 per call: 980 calls, 5.5M input and 1.5M output tokens for 97,931 edges, all of which received a label.

Synonym merging. To discount surface variation, all labels are embedded (`text-embedding-3-small`, 512 dimensions) and merged by leader clustering: in decreasing order of frequency, each label joins the most similar existing leader if the cosine similarity is at least t , and otherwise becomes a leader (`relation_vocab.py`). We report $t = 0.85$. At $t = 0.80$ the largest cluster already merges opposite relations (`plays_for` with `plays_against`), so 0.85 is the most aggressive threshold that still merges only synonyms; lower thresholds are in `relation_vocab.json`.

Direction. For a derived predicate, an edge is stored against its relation when the subject of the predicate is the edge’s target. We also report the fraction of predicates with at least 10 edges where at least 10% of the edges go each way. In WUKONG direction is fixed by the endpoint types of each relationship type. LightRAG stores an undirected graph.

Properties. Typed property values per node, and the fraction of nodes whose content is only free text (a description).

Control: recall against gold. Each gold row is matched to graph nodes by the article’s title and the gold name, after normalization (case, accents, punctuation), and by aliases: for people, the same surname with a compatible first name (prefix or common nickname); for teams, the nickname alone; for cities, *City*, *State* and *City of X*. WUKONG nodes must also have the right type; LightRAG and GraphRAG nodes may have any type. A gold attribute value (non-empty, and non-zero for counts) is found in WUKONG if the typed field holds it (numbers within 1%, city areas in either square miles or km², since the gold table mixes them). For LightRAG and GraphRAG it is found if it appears anywhere in the descriptions of the matched nodes or of their edges: dates in the usual written forms, counts only within 60 characters of a word naming what is counted, positions through *forward/center* and *guard*. This is an upper bound: the value is in the graph but not in a form a query can select. A gold relation (player’s team, team’s city, team’s owners, owner’s team) is found in WUKONG if an edge of the right type joins the matched nodes, and in LightRAG and GraphRAG if any edge joins them (`e1_analysis.py`).

4 Results

		WUKONG	LightRAG	GraphRAG
Size	Nodes	2,113	34,162	37,917
	Edges	3,376	44,040	53,891
Node vocabulary	Node types	4	14	99
	Node types used once	0	1	94
Relation vocabulary, own labels	Labels	3	25,151	none
	Labels used once	0%	79%	–
	Labels to cover 90% of edges	1	20,747	–
Relation vocabulary, derived predicates	Labels	3	4,388	4,428
	Labels used once	0%	56%	52%
	Labels to cover 50% / 90% of edges	1 / 1	51 / 1,079	47 / 936
	Labels per node-type pair (mean)	1	310	736
Synonyms merged (cos ≥ 0.85)	Labels	3	3,488	3,477
	Labels to cover 90% of edges	1	738	633
Direction	Edges stored against their relation	0%	48% ^a	16%
	Frequent labels used in both directions	0%	85%	56%
Properties	Typed property values per node	1.5	0	0
	Nodes described only by free text	0%	100%	100%
Control: recall vs. gold	Entities found	214/216	214/216	214/216
	Attribute values found ^b	859/955	714/955	729/955
	Relations found ^c	172/220	195/220	192/220

Table 2: Structure of the three graphs. For WUKONG the derived and merged rows repeat its own labels. ^aLightRAG’s graph is undirected. ^bFor LightRAG and GraphRAG, a value counts if it appears in their text (upper bound). ^cFor LightRAG and GraphRAG, any edge between the two entities counts, whatever relation it describes; most of their extra relations are players’ teams outside the NBA, which the WUKONG schema does not target.

Notes.

- The three graphs find the same entities and comparable numbers of facts. WUKONG finds more attribute values as typed fields than the baselines hold anywhere in their text; the baselines connect more player–team pairs, mostly with non-NBA clubs.
- In the derived labelings, about 50 predicates cover half of the edges and about 1,000 are needed for 90%; more than half of all predicates label a single edge. Merging synonyms at $t = 0.85$ reduces the 90% figure to 738 and 633.
- Example: in LightRAG, the 892 edges between a player and a team in the gold tables carry 676 distinct keyword labels; *played for* labels 17 of them.

Limitations. Each system was run once. As an indication of run-to-run variation, three runs of each of two WUKONG schema variants close to `workspace-v1` differ by at most 3 facts in any row of the gold comparison, 9 attribute values and 3 relations in total, 25 nodes and 76 edges. Derived predicates come from labeling each edge independently, without a shared vocabulary, which is the situation of a user who wants to query these graphs by relation. The gold tables contain errors (mixed units for city areas, legal names for owners), which affect all systems alike.

Reproduction

From `paper/uda-nba/`: `run_full.sh` builds the baseline graphs, `run_variants.sh` builds WUKONG with the clean schemas, then `e1_analysis.py`, `derive_labels.py` (once per baseline), `relation_vocab.py` and `e1_table.py`, which writes `runs/e1/summary.csv`. Data, environments and runs are not versioned.